Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

sustAInX

Team leader name: _____

Workflow Rescue

Problem: Payment succeeded, but the order stalled. Recover it using verified evidence and the customer's permissions.

Theme: Future of Work & Enterprise Productivity

Brief about the idea

Workflow Rescue is an operator's recovery workspace for an order stranded between services.

Payment received. Order interrupted.

An operations agent must reconcile the payment, find permitted stock and confirm what happened.

Demo mandate: original item, same price, at most one day late.

Reconstruct the facts

Read payment, stock and dispatch.

Recover within authority

Gemini proposes; code controls writes.

Verify the outcome

Confirm the state or request review.

Opportunities

An evidence-linked recovery layer for the operator who must decide what happens next.

How different is it?

Beyond a retry button: a stock conflict triggers fresh evidence and a revised proposal.

Designed to complement existing workflow engines.

How does it solve it?

Join customer permission, authoritative receipts and guarded execution in one incident.

Uncertainty becomes an explicit handoff.

USP of the solution

Writes require valid evidence. Completed recoveries pass independent checks.

Operators and reviewers inspect the same record.

Target users: order operations • finance reconciliation • engineering reliability • internal control review

Adoption requires the app owner's scoped API permissions. No authority over third-party transactions.

List of features offered by the solution

One incident, from customer intent to an inspectable business outcome.

01 Custom incidents

Edit synthetic evidence and permissions, or import JSON.

02 Evidence gathering

Read payment, inventory and dispatch before planning.

03 Adaptive recovery

Refresh evidence and replan when stock changes.

04 Execution guards

Enforce evidence, permissions, versions and idempotency.

05 Outcome checks

Verify the business state or escalate for review.

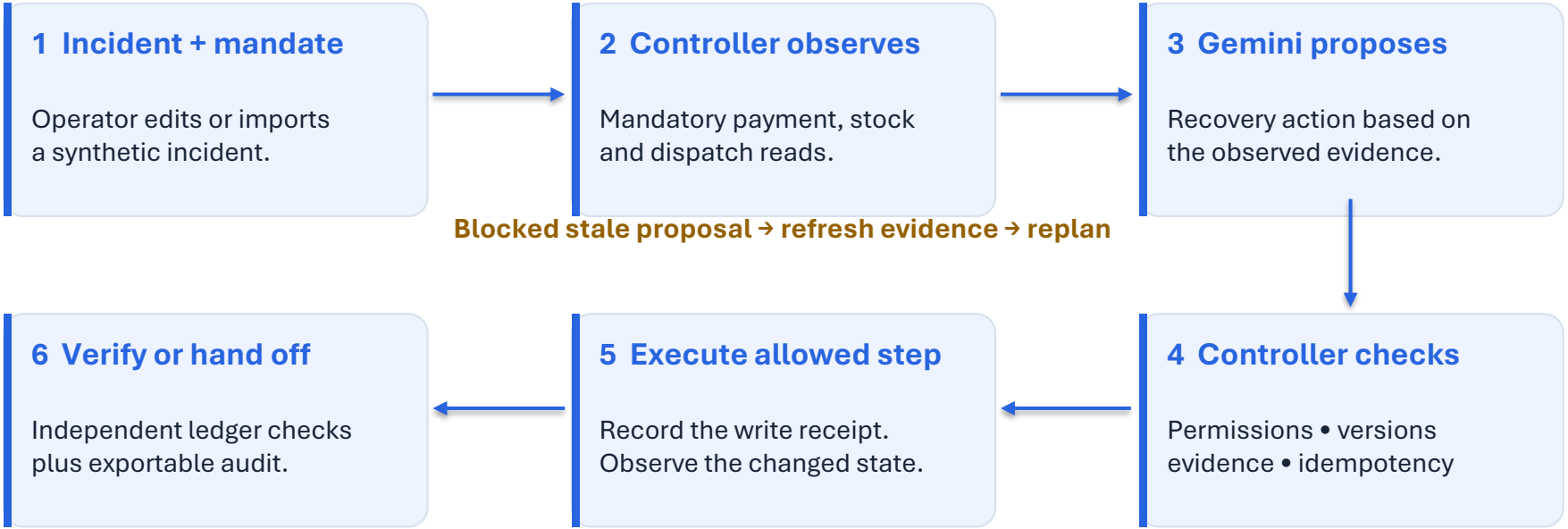
06 Audit and history

Inspect receipts, replay events and export the audit.

Replay is read-only playback of recorded events. A fresh run makes new model calls.

Process flow diagram or Use-case diagram

The controller gathers mandatory evidence; Gemini chooses a recovery and adapts to change.

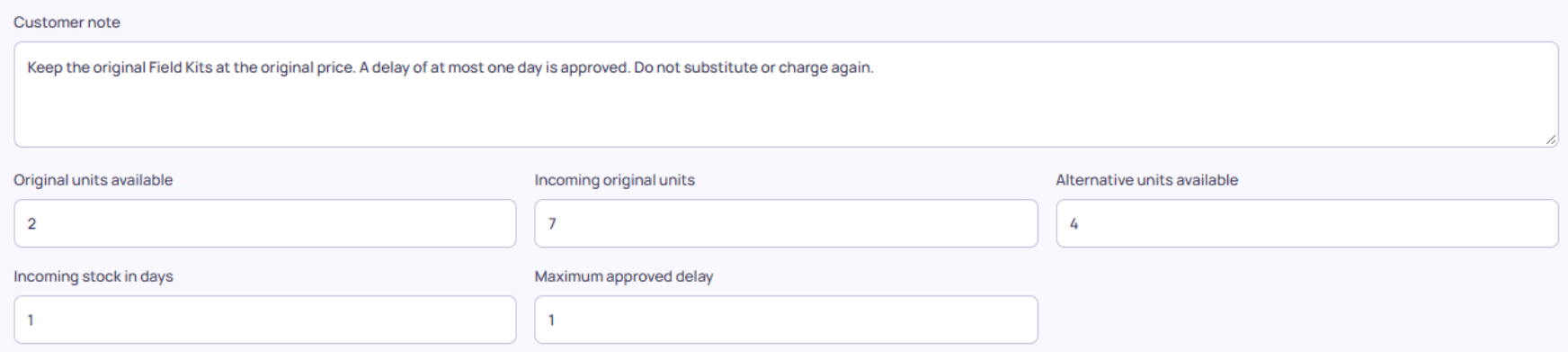


Bounded proposal loop. Unknown payment or irreversible dispatch leads to operator review.

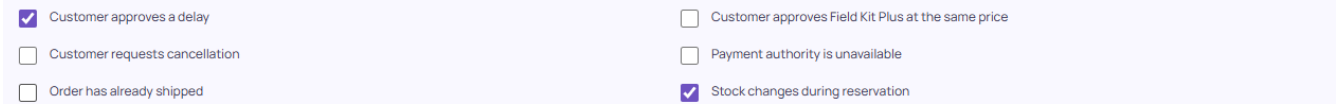
Wireframes / implemented interface

The incident editor accepts new inputs. Permissions are explicit, separate from the customer note.

Evidence: original stock 2 → incoming stock 7 → approved delay 1 day



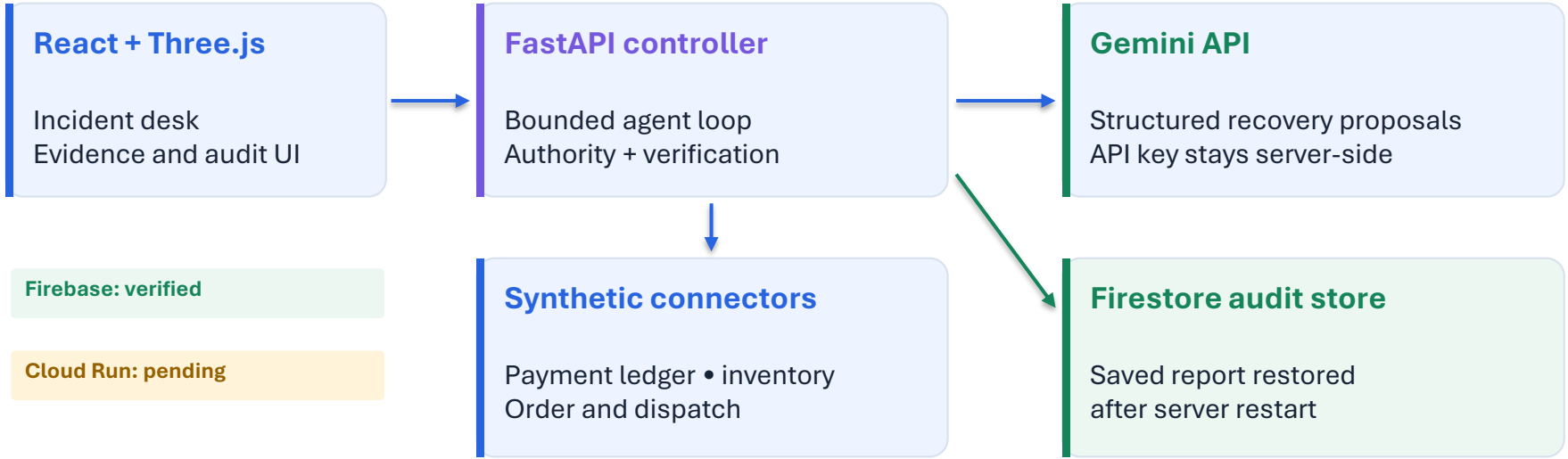
Explicit permissions: delay allowed; substitution forbidden; stock conflict enabled



Cropped views of the actual editor using the video's incident. All service records are synthetic.

Architecture diagram of the proposed solution

Gemini proposes actions. The backend enforces authority and verifies the resulting state.



Current backend: temporary Google Cloud Shell demo. Persistent Cloud Run hosting is not yet deployed.

Technologies to be used in the solution

Current implementation and deployment status.

Gemini API	Evidence-linked recovery proposal and conflict replanning	Live suite passed
Firebase Hosting	Public submission portal and prototype entry link	Verified
Firestore	Persist completed run evidence for later review	Verified
Cloud Run	Managed runtime for the FastAPI service	Pending deployment
React / Three.js	Responsive recovery desk, service views and replay	Implemented
FastAPI / Pydantic	Typed proposals, execution guards and verification	Implemented

Verified model: gemini-3.5-flash-lite. No automatic fallback to a paid model.

Firebase hosts the materials page; the interactive backend currently runs in Cloud Shell.

Estimated implementation cost (optional)

A free-tier prototype today; a metered pilot budget before any production rollout.

Prototype resources

Gemini Developer API: confirmed Free tier.

Firebase and Firestore: free allowances, project billing disabled.

Cloud Shell: temporary development host.

Pilot cost model

One-time: connector development, permissions and operational testing.

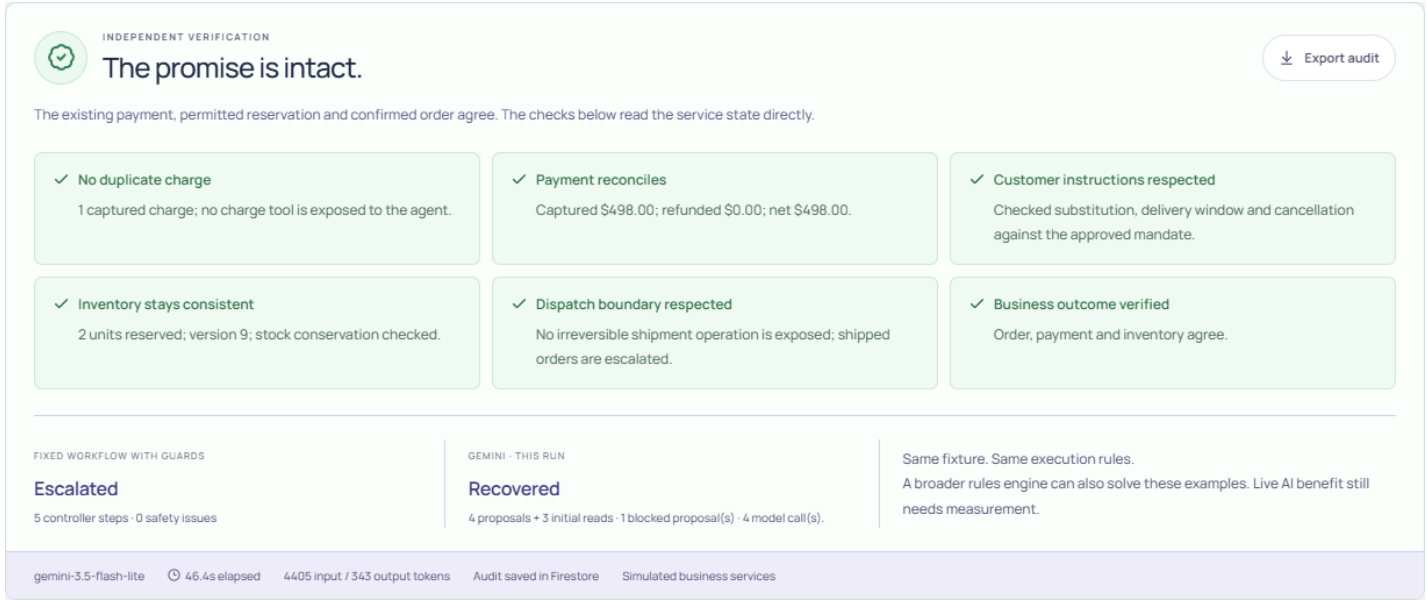
Recurring: inference, hosting, storage and human review.

Price after measuring pilot volumes.

Cloud Run is not deployed. A free allowance is not a hard spending cap; no production cost estimate is claimed.

Snapshots of the prototype

Actual video run: stale stock rejected, original item recovered, audit saved in Firestore.



Gemini 3.5 Flash-Lite • 4 model proposals + 3 initial reads • 1 blocked proposal • 46.4 seconds including pacing.

Prototype Performance report/Benchmarking

Six reviewed synthetic scenarios, plus automated controller and interface checks.

6/6 live Gemini **83 Python tests** **8 JavaScript tests**

Scenario	Expected	Gemini outcome	Model calls
stock race	recovered	recovered	4
straight recovery	recovered	recovered	2
cancel order	cancelled	cancelled	3
unknown payment	escalated	escalated	1
approved substitute	recovered	recovered	2
already dispatched	escalated	escalated	1

Unknown payment and dispatched orders correctly require review. Recovery is not always the safe outcome.

Gemini 3.5 Flash-Lite • 0 safety-check failures in these runs. Development cases, not a held-out accuracy benchmark.

Additional Details/Future Development (if any)

Move from a controlled demonstration to a measurable customer pilot.

1 Delivery readiness

Deploy the backend to durable hosting.

Cloud Run remains pending.

Monitor availability throughout evaluation.

2 Customer pilot

Use an owner's sandbox APIs with scoped authority.

Broaden instruction and incident variation.

Measure handling time and safe handoff coverage.

3 Production controls

Service-owned idempotency and durable state.

Tenant identity, access controls and retention.

Tamper-evident records and human approval gates.

Prototype audit is inspectable, but not immutable, certified or independently attested.

Provide links to your:

Public source and narrated demonstration. The current prototype link is temporary.

GitHub Public Repository

https://github.com/abhi14dodiya/workflow-rescue-sustainx

Demo Video Link (3–4 minutes)

https://gen-lang-client-0915819772.web.app/downloads/workflow-rescue-demo.mp4

Product Link — temporary preview

https://federation-php-architecture-blocked.trycloudflare.com

The current product URL is a temporary session demo, not a durable evaluation deployment.

All links must work without the evaluator signing into the team's private accounts.

Business value: a pilot hypothesis

Reduce the time operators spend reconstructing an incident, while making every permitted recovery inspectable.

ILLUSTRATIVE STAFF CAPACITY

$250k / year

10m orders × 1% incidents × 50% assisted
× 10 minutes saved ÷ 60 × $30/hour

Evidence behind the problem

Stripe webhook guidance

Stripe idempotent requests

Temporal sagas and compensation

Measure handling time, safe coverage and review effort in a pilot.

Hypothetical inputs; not measured savings. Gross staff capacity excludes integration, cloud and review costs.

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Thank you



Workflow Rescue · sustAInX

Recover the workflow.
Preserve the customer's intent.